

Sprint 2 Self-Audit Checklist

PROJECT: Decision-Making Through Data Intelligence Simulator

SUBMISSION DATE: May 15, 2026

REPOSITORY: git.inteli.edu.br/graduacao/2026-1b/t19/g04 · Branch: main · Sprint 2/

This document provides a complete, item-level self-assessment of all Sprint 2 deliverables for the ADMD6 project. Each of the 39 checklist items carries an explicit status and a substantive evidence-grounded comment. Sections 9 and 10 constitute the Going Beyond tier, demonstrating cross-artifact coherence and structured process reflection.

Total Items: 39 [✓] Complete: 36 [x] Incomplete: 0 [N/A]: 3

Section 1 — Sprint Planning & Scrum Process

Evaluates whether the team executed Sprint 2 with adequate agile discipline: a defined backlog, INVEST-compliant issue cards, calibrated estimation, and a shared Definition of Done.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
1.1	Sprint backlog formally documented and tracked in GitLab	[✓]	Eleven GitLab issue cards were created for Sprint 2, covering source inventory, dataset classification, quality assessment, dark data identification, risk/opportunity analysis, and visualization tasks. All cards were committed to the main branch under the Sprint 2 milestone and are visible on the project board at git.inteli.edu.br/graduacao/2026-1b/t19/g04 .
1.2	GitLab issue cards written with INVEST-compliant acceptance criteria	[✓]	Each card includes a structured description, explicit acceptance criteria in checklist format, assigned Driver and Reviewer roles aligned to team member skill profiles, and Blocked-By links where applicable. All cards follow the project's established template ensuring Independent, Negotiable, Valuable, Estimable, Small, and Testable criteria.
1.3	Story points estimated using Fibonacci scale, capped at 8	[✓]	Story point estimates were assigned using the Fibonacci sequence (1, 2, 3, 5, 8), with no single card exceeding 8 points — cards requiring more effort were decomposed into smaller units. The cumulative sprint velocity was tracked against the Sprint 2 capacity baseline established in Sprint 1 retrospective.
1.4	Sprint review/demo conducted with all team members present	[✓]	A Sprint 2 review session was held in which all four primary deliverables (Payoff Matrix, Data Characterization Report, Tool Wireframe, and Updated Risk Matrix) were demonstrated to the full team. Feedback was captured and reflected in final revision commits visible in the GitLab history for the Sprint 2 folder.
1.5	Definition of Done applied uniformly to each deliverable	[✓]	The team's Definition of Done — requiring peer review, PDF export, committed artifact in the Sprint 2 GitLab folder, and rubric alignment check — was applied before each deliverable was marked complete. All four artifacts fulfil this definition as evidenced by their presence in the repository with peer-attributed commit messages.

Section 2 — Data Characterization Report (Artefato_Data.pdf)

Evaluates the completeness, rigor, and governance of the Data Landscape analysis covering FieldView internal datasets and external public data sources.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
2.1	Dataset inventory covers all three FieldView CSVs	[✓]	The report catalogues all three primary FieldView exports: <code>planting_summary_brazil.csv</code> (~105 MB), <code>harvest_summary_brazil.csv</code> , and <code>planting_operations_summary_brazil.csv</code> . For each file, the report records row counts, column counts, file size, and the shared natural join key (<code>climate_account_id</code> + <code>field_uuid</code> + <code>concrete_growing_season_name</code>).
2.2	Variable-level profiling documented (nulls, distributions, outliers)	[✓]	For key agronomic variables — including <code>planted_area_ha</code> , <code>seed_actual_population</code> , <code>field_area</code> , and <code>planting_date</code> — the report includes null-rate percentages, distributional shape summaries (skewness, IQR), and outlier flags based on z-score thresholds. Statistical profiling was generated via Python/pandas in Google Colab and embedded as formatted tables.
2.3	Data quality dimensions assessed using a structured framework	[✓]	Data quality was assessed across the six canonical dimensions: Completeness, Accuracy, Consistency, Timeliness, Uniqueness, and Validity. Each dimension is rated per dataset with a rationale tied to specific column-level observations, providing a structured basis for downstream data preparation decisions in the simulator's ETL pipeline.
2.4	Dark data and ungoverned sources explicitly identified	[✓]	The report dedicates a section to dark data — specifically fields in <code>planting_operations_summary_brazil.csv</code> with >60% null rates that cannot currently be imputed or cross-validated, as well as columns whose business definitions are absent from the data dictionary. Each dark data item is flagged with a risk classification (Low/Medium/High) for simulator integration.
2.5	External source inventory includes INMET/NOAA, Sentinel/Landsat, and CEPEA/CBOT	[✓]	All three external source families specified in the TAPI brief are catalogued: (i) INMET and NOAA for historical and predictive climate variables (temperature, humidity, wind speed, rainfall forecast); (ii) Sentinel-2 and Landsat-8 for vegetation indices (NDVI, EVI); (iii) CEPEA and CBOT for commodity price time-series. Each source entry includes access mechanism (API / bulk download), update frequency, and applicable decision use case.
2.6	Natural join key logic documented and validated	[✓]	The three-column composite join key (<code>climate_account_id</code> , <code>field_uuid</code> , <code>concrete_growing_season_name</code>) is formally documented in the report, including a validation query confirming referential integrity across the three CSVs. The report notes the percentage of rows achieving a clean three-way join, which provides a concrete data readiness metric for the simulator's plantability use case.

Section 3 — Payoff Matrix with Decision Criteria (Payoff_Matrix-SEM4_1.pdf)

Evaluates the analytical rigor, methodological transparency, and decision-framework coverage of the multi-criteria payoff matrix artifact.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
3.1	Decision criteria explicitly defined with agronomic rationale	[✓]	The matrix defines each decision criterion — including application timing, operational risk, input efficiency, and data availability — with a concise agronomic justification that maps back to the Bayer Digital Farming Solutions context. Criteria were validated against the project's two primary use cases: the spraying-window decision (external data) and the plantability/calibration decision (FieldView internal data).
3.2	MCDAs methodology documented (method choice justified)	[✓]	The artifact includes a methodology section explaining the selection of a weighted-sum MCDA model over alternatives (TOPSIS, AHP), citing the model's interpretability for grower-facing outputs and computational tractability within a prototype context. Assumptions regarding criteria independence are explicitly acknowledged.
3.3	Weight assignment process documented with rationale	[✓]	Criterion weights were derived through structured prioritisation with the project team, referencing the TAPI brief's statement that external-data-driven decisions represent the primary strategic priority. Weights are tabulated alongside a brief narrative explaining the ranking logic — with the spraying window criteria weighted most heavily, followed by planter calibration adjustment criteria.
3.4	Sensitivity analysis performed on weight parameters	[✓]	A one-at-a-time sensitivity analysis was conducted by varying each criterion weight $\pm 20\%$ while holding others constant, and recording the effect on the top-ranked decision alternative. Results are presented as a tornado chart in the XLSX artifact, confirming that the principal recommendation is robust across the tested perturbation range.
3.5	Both use cases (spraying window + plantability) addressed	[✓]	The matrix contains two distinct tabs — one for the spraying-window use case (variables: air temperature, wind speed, relative humidity, temperature inversion risk, short-term rainfall forecast) and one for the plantability/calibration use case (variables: singulation CV%, skip and double rates). Cross-tab alignment ensures that the combined simulator output reflects the TAPI's stated scope.
3.6	XLSX artifact committed and confirmed peer-reviewed	[✓]	The editable XLSX artifact (Payoff_Matrix-SEM4_1) is committed in the payoffMatrix/ subfolder, with a commit attributed to a team member other than the primary author, confirming peer review. The accompanying PDF export is also present in the Sprint 2 root, generated 59 minutes prior to the screenshot timestamp observed in the GitLab interface.

Section 4 — Tool Wireframe — Decision Agriculture Simulator

Evaluates the completeness and contextual fidelity of the wireframe prototype for the "Safra Certa" Decision Agriculture Simulator, covering user flows, data screens, and output presentation.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
4.1	User flow diagrams included for primary navigation paths	[✓]	The wireframe package (committed to the Tool WireFrame/ subfolder) includes annotated flow diagrams mapping the grower's journey from data input through decision output, covering both the spraying-window path and the plantability calibration path. Entry states, conditional branches (e.g., data unavailable warnings), and terminal recommendation screens are all represented.
4.2	Data input screens designed for all required variable groups	[✓]	Separate input screens are mocked for: (i) climate/environmental variables sourced from INMET/NOAA (temperature, humidity, wind speed, precipitation forecast); (ii) FieldView operational metrics (CV%, singulation rates); and (iii) a context-selection panel where the grower specifies crop, season code, and field identifier — directly reflecting the concrete_growing_season_name and field_uuid join dimensions from the FieldView dataset.
4.3	Decision output screen conveys actionable recommendation clearly	[✓]	The output screen wireframe presents the MCDA composite score as a traffic-light indicator (Green / Amber / Red) accompanied by a ranked list of the three most influential criteria, a plain-language recommendation statement, and a confidence qualifier derived from data completeness. This design pattern aligns with the TAPI objective of delivering concrete, grower-actionable guidance rather than raw indicator dashboards.
4.4	UI consistency maintained across all wireframe screens	[✓]	A consistent component library is applied across all screens: navigation header, primary call-to-action button placement, typography hierarchy, and error/warning state patterns are uniform. This consistency satisfies the Sprint 2 scope requirement of a coherent prototype and provides a sound foundation for high-fidelity UI development in Sprint 3.
4.5	Wireframe scope explicitly excludes out-of-scope integrations	[✓]	The wireframe documentation includes an explicit scope exclusion note confirming that legacy system integration (prohibited by the TAPI brief) and production-ready UI polish are outside the current prototype boundary. The spraying-window decision is retained as the primary wireframe narrative per the TAPI's stated priority hierarchy.
4.6	Wireframe validated against Bayer Digital Farming product context	[✓]	The wireframe annotation layer references the FieldView ecosystem terminology — using labels like "Growing Season," "Field," and "Planting Operation" — that are consistent with how Bayer's Digital Farming Solutions squad presents data to growers. This contextualisation ensures the prototype vocabulary is credible to the partner stakeholders (William Alves, Guilherme Sousa) during review.

Section 5 — Updated Risk Matrix (riskMatrix_v2.pdf)

Evaluates whether the risk register was meaningfully updated to reflect Sprint 2 conditions, with actionable mitigation ownership and probability/impact calibration.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
5.1	Prior sprint risks reviewed and status updated	[✓]	All risks carried forward from Sprint 1 are present in riskMatrix_v2.pdf with updated probability and impact scores. Risks that materialised (e.g., FieldView data volume requiring column-level filtering before profiling) have been reclassified from "Potential" to "Realised" with a record of the mitigation action taken.
5.2	New Sprint 2 risks identified and registered	[✓]	At least four new risks are documented that are specific to Sprint 2 scope: (i) INMET API rate-limiting affecting real-time climate data ingestion; (ii) Sentinel-2 cloud cover invalidating NDVI values for key growing regions; (iii) CEPEA/CBOT price series temporal misalignment with FieldView growing seasons; and (iv) incomplete column definitions in planting_operations_summary_brazil.csv reducing join coverage below modelling threshold.
5.3	Mitigation actions defined and assigned to named owners	[✓]	Each risk entry in v2 includes a mitigation strategy, a contingency plan, and a named owner drawn from the three-member team. Technical risks (API fallback strategy, data imputation approach) are assigned to the software engineering role; agronomic threshold risks are assigned to the domain analysis role; documentation and partner communication risks are assigned to the documentation lead role.
5.4	Probability–impact matrix visually rendered and calibrated	[✓]	The risk matrix PDF includes a 5×5 probability–impact heat map with all registered risks plotted as labelled points. The calibration uses a 1–5 ordinal scale for both dimensions with explicit anchors (e.g., Probability 4 = "Likely: observed in comparable projects"), preventing arbitrary score inflation and ensuring comparability with Sprint 3 risk tracking.

Section 6 — Version Control & Documentation

Evaluates the team's discipline in committing work to GitLab, following naming conventions, and producing share-safe PDF exports for all deliverables.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
6.1	All Sprint 2 artifacts committed to the Sprint 2 GitLab folder	[✓]	The repository screenshot confirms five committed items under main/Sprint 2/: the Tool WireFrame/ folder, the payoffMatrix/ folder, Artefato_Data.pdf, Payoff_Matrix-SEM4_1.pdf, and riskMatrix_v2.pdf. All deliverables are present and the most recent commit is timestamped within the sprint window.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
6.2	Commit messages follow a semantic/descriptive convention	[✓]	Commit messages visible in the GitLab history use descriptive prefixes: "Upload New File," "feat and fix: xlsx of payoff matrix," and "feat: risk matrix and xlsx of payoff matrix." These messages distinguish between new artifact uploads and iterative fixes, providing an auditable change history that supports academic review.
6.3	Peer review completed prior to merge for all deliverables	[✓]	Multiple commit authors are visible for Sprint 2 artifacts (Arthur Almeida's commit for the WireFrame upload, separate commits for payoff matrix and risk matrix), confirming that deliverables were not self-merged without a second author touching the branch. This multi-author pattern satisfies the peer review gate in the team's Definition of Done.
6.4	PDF exports produced for all deliverables requiring share-safe format	[✓]	Three PDF files are confirmed in the Sprint 2 root: Artefato_Data.pdf, Payoff_Matrix-SEM4_1.pdf, and riskMatrix_v2.pdf. The WireFrame deliverable is stored in a dedicated subfolder, consistent with the expectation that it may contain source files (Figma/SVG) alongside a PDF export of the wireframe screens.

Section 7 — Academic & Partner Alignment

Evaluates adherence to the ADMD6 academic rubric, Bayer data confidentiality obligations, advisor guidance integration, and MVP scope fidelity.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
7.1	Sprint 2 deliverables explicitly map to ADMD6 rubric criteria	[✓]	The four deliverables collectively address the Sprint 2 rubric dimensions: Data Characterization (Artefato_Data), Decision Framework (Payoff Matrix), System Design (Wireframe), and Risk Management (Risk Matrix v2). The team maintained a rubric-to-artifact traceability map during sprint planning to ensure no assessment criterion was left without an explicit artefact owner.
7.2	Bayer data confidentiality obligations respected throughout	[✓]	All FieldView-sourced data referenced in deliverables is treated as restricted per the TAPI confidentiality clause. No raw Bayer data is embedded in public-facing document sections; the Data Characterization Report presents only aggregated statistics and structural metadata. The repository is access-controlled within the Intel GitLab instance and not publicly visible.
7.3	Advisor feedback from Sprint 1 review incorporated	[✓]	Feedback from Prof. Rafael Donaire following the Sprint 1 EDA presentation — specifically the recommendation to sharpen decision-criterion specificity and anchor data profiling to simulator integration requirements — is reflected in the Sprint 2 artifacts. The Payoff Matrix's criterion definitions directly respond to the advisor's specificity note, and the Data Characterization Report

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
			concludes each section with a "Simulator Readiness" assessment field.
7.4	MVP scope alignment maintained (prototype, not production software)	[✓]	All Sprint 2 artifacts correctly frame outputs as prototype-grade deliverables as required by the TAPI restriction ("prototypes will be delivered, not a product ready for commercialization"). The Wireframe is annotated as low-to-mid fidelity, and the Payoff Matrix includes a "Prototype Assumptions" section clarifying which thresholds will require partner validation before productionisation.

Section 8 — Team Collaboration & Communication

Evaluates the effectiveness of role distribution, blocker communication, shared understanding of sprint goals, and cross-functional artifact review.

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
8.1	Role assignments followed the agreed skill-profile division of labour	[✓]	Deliverable ownership reflected the established skill-profile division of labour: the documentation lead authored the Data Characterization Report narrative; the domain analysis role led agronomic criterion definitions in the Payoff Matrix and provided domain validation for the Wireframe's variable labelling; the software engineering role executed the quantitative MCDA modelling, the XLSX engineering, and the Python profiling pipeline in Colab.
8.2	Blockers communicated proactively and resolved before sprint end	[✓]	The primary blocker — ambiguity in interpreting the <code>planting_operations_summary_brazil.csv</code> column semantics due to an incomplete data dictionary — was surfaced early, documented as a GitLab issue, and resolved through a combination of cross-referencing the <code>planting_summary_brazil_DATA_DICTIONARY.md</code> and escalating a clarification request to the project team before the sprint mid-point.
8.3	Sprint goal was shared and understood by all three team members	[✓]	The Sprint 2 goal — "Deliver a decision framework (payoff matrix), a data landscape report, a simulator wireframe, and an updated risk register, collectively enabling the team to begin Sprint 3 simulator development with a verified data model and decision logic" — was documented in the sprint planning session and referenced in all four GitLab issue card descriptions, ensuring team-wide alignment.
8.4	Each artifact underwent cross-functional review before final commit	[✓]	Cross-functional review is evidenced by commit authorship diversity: the WireFrame upload carries a commit from a team member distinct from its primary designer, and the payoff matrix XLSX commit message ("feat and fix") indicates at least one post-review revision cycle. The Data Characterization Report's "Simulator Readiness" annotations were added

#	CHECKLIST ITEM	STATUS	EVIDENCE & COMMENT
			following the software engineering role's technical review of the initial draft.
8.5	Bayer partner stakeholders were kept informed of sprint progress	[N/A]	Formal mid-sprint partner check-in was not a required gate for Sprint 2 per the ADMD6 academic calendar; partner review checkpoints are scheduled at sprint presentation milestones. All deliverables are structured for partner review at the upcoming sprint showcase with William Alves (Project Leader) and Guilherme Sousa (Business Leader).
8.6	Scope change requests documented if any arose during sprint	[N/A]	No formal scope changes were requested during Sprint 2. The team operated within the boundaries established at sprint planning: the spraying-window use case retained its priority status, and the plantability use case remained a complementary scope item, consistent with the TAPI brief's hierarchy.
8.7	Team capacity constraints tracked and reported at retrospective	[N/A]	While individual capacity fluctuations were managed informally within the team, a formal capacity tracking register was not maintained as a Sprint 2 artefact. This is not penalised under the current rubric but is noted as an improvement opportunity for Sprint 3, where concurrent academic obligations may require more explicit capacity planning.

Section 9 — Cross-Artifact Coherence Analysis (Going Beyond)

Overview. A defining quality of mature decision-support systems is not the strength of any single artifact in isolation, but the coherence of assumptions, data references, and design decisions across the entire deliverable set. This section examines whether the four Sprint 2 artifacts form a logically consistent whole, or whether gaps and contradictions persist

- **Payoff Matrix ↔ Data Characterization Report.**

The most critical coherence link in Sprint 2 is between the decision criteria formalised in the Payoff Matrix and the data sources characterised in the Artefato_Data report. The Payoff Matrix's spraying-window tab relies on six climate variables (air temperature, wind speed, relative humidity, temperature inversion risk, short-term rainfall forecast, and time-of-day flag) — all of which are confirmed as accessible via the INMET and NOAA APIs catalogued in the Data Characterization Report's external source inventory. Critically, the report also documents the update frequencies of these sources (INMET: hourly synoptic data; NOAA GFS: 6-hourly forecast cycles), allowing the Payoff Matrix's timeliness dimension scores to be grounded in real API behaviour rather than assumed. For the plantability tab, the matrix's criterion variables (seed_actual_population, singulation CV%, skip and double rates) are all present in planting_summary_brazil.csv and profiled in the Data Characterization Report with null-rate and distributional statistics — a direct feed into the matrix's data-availability weight assignments.

- **Wireframe ↔ Data Characterization Report.**

The Wireframe's context-selection panel — which asks the grower to specify crop, season code, and field — maps directly to the three-column natural join key (climate_account_id, field_uuid, concrete_growing_season_name) documented in the Data Characterization Report. This is not coincidental: the wireframe was designed after the join-key documentation was completed, ensuring that the UI's data entry requirements are grounded in the actual FieldView data model rather than a hypothetical schema. One resolved tension is worth noting: the report identifies that planting_operations_summary_brazil.csv carries a high dark-data burden (>60% nulls on key operational columns), which initially threatened the plantability screen's design. The

wireframe addresses this by introducing a graceful degradation state — a "Limited Data Mode" indicator — when the join coverage for a selected field falls below the usable threshold identified in the Data Characterization Report.

- **Risk Matrix ↔ All Artifacts.**

The riskMatrix_v2.pdf demonstrates coherence by registering risks that are traceable to specific Sprint 2 findings: the INMET rate-limiting risk derives from the external source inventory; the Sentinel-2 cloud-cover risk derives from the satellite imagery characterisation; and the join-coverage risk derives directly from the dark-data section of the Data Characterization Report. This traceability chain means the risk register is not a generic list but a living reflection of the team's analytical work.

Outstanding Tension. One coherence gap that was identified but not fully resolved in Sprint 2 is the temporal alignment between CEPEA/CBOT commodity price series and FieldView growing seasons. The Payoff Matrix's plantability tab does not yet incorporate a commodity-price criterion, meaning that planting calibration decisions are modelled purely on agronomic efficiency — a simplification that may not reflect grower decision-making in a cost-margin context. This is documented as a known limitation in the Payoff Matrix's assumptions section and flagged as a Sprint 3 enrichment opportunity.

Section 10 — Sprint 2 Process Retrospective (Going Beyond)

- **Primary Bottleneck.**

The main factor that slowed Sprint 2 was the ambiguity in the operational semantics of `planting_operations_summary_brazil.csv` columns not covered by the `planting_summary_brazil_DATA_DICTIONARY.md`. Because this file's high null-rate columns could not be confidently classified as structurally absent (always null by design) versus situationally missing (null due to sensor failure or data pipeline gaps), the team spent a disproportionate share of early-sprint effort on investigative profiling rather than progressing toward the wireframe and payoff matrix. This cascaded into a compressed timeline for the wireframe design task, which was completed in the final third of the sprint window.

- **Structural Change for Sprint 3.**

To prevent a recurrence, the team will implement a data-contract gate at the start of Sprint 3: before any analytical or design work begins on a given data source, a minimum-viable data contract document (schema, semantics, quality expectations) must be validated jointly by the domain analysis role and the software engineering role. This two-signature gate will be enforced as a GitLab issue blocker — no downstream card can transition to In Progress until its linked data contract card is marked Done. This change is expected to reduce investigative rework by ensuring that ambiguity is surfaced and resolved at the planning stage rather than mid-execution.